

## Module 3 Summary – Jacob booth 221530207

### Response to feedback

Regarding feedback for this module, ensuring my summary:

- Answers all guiding questions
- Shows clear connections between the concepts
- Is written in paragraph form
- Highlights key terms in bold to add extra structure

My last attempt was too modular and too verbose. It was helpful for mapping out the links but wasn't refined enough to convey the ideas cohesively. I've changed the structure to paragraph form, emphasising concept-integration so each topic is discussed in context, with clearer key-takeaways and omitting rephrasing. I've maintained my emphasis on concept inter-dependence and addressed the guiding questions more directly. This attempt focuses on gradually expanding the overall picture as opposed to phrasing and linking each definition/theorem individually.

### Module Learning Objectives

I certify that I achieved the following learning objectives for the module:

- 1) Apply the inversion algorithm to find a matrix inverse
- 2) Identify and work with elementary matrices
- 3) Work with matrices algebraically and identify correspondence between key linear mappings and their matrices

### Key Definitions and Theorems

#### Matrix Inverses

| Textbook Ref                                  | Theorem/Definition statement                                                                                                                                                                                                        | Learning Notes                                                                                                                                                                                        |
|-----------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Definition 2.11</b><br>Matrix Inverse      | <p>If A is a square matrix, a matrix B is called an <u>inverse</u> of A if and only if</p> $AB = I \text{ and } BA = I$ <p>A matrix A that has an inverse is called an <u>invertible matrix</u> and denoted <math>A^{-1}</math></p> | DIFFERENT FROM THE NEGATION, remember that this isn't the negative matrix but its inversion                                                                                                           |
| <b>Theorem 2.4.1</b><br>Uniqueness of inverse | If B and C are both inverses of A, then $B = C$ .                                                                                                                                                                                   | Think about the rules with matrix multiplication, sometimes C might not be known but must still be identical if it returns the same result as B, there's a single inverse for every invertible matrix |

|                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                          |
|--------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Matrix Inversion Algorithm</b>          | <p>If A is an invertible (square) matrix, there exists a sequence of elementary row operations that carry A to the identity matrix I of the same size, written <math>A \rightarrow I</math>. This same series of row operations carries I to <math>A^{-1}</math>; that is, <math>I \rightarrow A^{-1}</math>. The algorithm can be summarized as follows:</p> $[A \mid I] \rightarrow [I \mid A^{-1}]$ <p>where the row operations on A and I are carried out simultaneously.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Note that the inverse doesn't always exist, and REMEMBER that the identity matrix is the matrix representation of scalar 1, it leaves it unchanged (replacing the 1s with any real x will be that scalar's matrix representation, I assume)                                                                              |
| <b>Theorem 2.4.4</b><br>Inverse properties | <p>All the following matrices are square matrices of the same size.</p> <ol style="list-style-type: none"> <li>1. I is invertible and <math>I^{-1} = I</math>.</li> <li>2. If A is invertible, so is <math>A^{-1}</math>, and <math>(A^{-1})^{-1} = A</math>.</li> <li>3. If A and B are invertible, so is AB, and <math display="block">(AB)^{-1} = B^{-1}A^{-1}.</math> </li> <li>4. If <math>A_1, A_2, \dots, A_k</math> are all invertible, so is their product <math>A_1A_2 \dots A_k</math>, and <math display="block">(A_1A_2 \dots A_k)^{-1} = A_k^{-1} \dots A_2^{-1}A_1^{-1}</math> </li> <li>5. If A is invertible, so is <math>A^k</math> for any <math>k \geq 1</math>, and <math display="block">(A^k)^{-1} = (A^{-1})^k.</math> </li> <li>6. If A is invertible and <math>a \neq 0</math> is a number, then <math>aA</math> is invertible and <math display="block">(aA)^{-1} = \frac{1}{a}A^{-1}</math> </li> <li>7. If A is invertible, so is its transpose <math>A^T</math>, and <math display="block">(A^T)^{-1} = (A^{-1})^T.</math> </li> </ol> | Note the swapping behaviour in the definition, applying inverses to the matrices inverts their corresponding multiplication order. If multiplying by the inverse returns "1" then it might be similar to multiplying the matrix by $1/A$ , or dividing $A/A$ ? Multiplying by the inverse seems to be the tool for this. |
| <b>Theorem 2.4.5</b><br>Inverse Theorem    | <p>The following conditions are equivalent for an <math>n \times n</math> matrix A:</p> <ol style="list-style-type: none"> <li>1. A is invertible.</li> <li>2. The homogeneous system <math>A\mathbf{x} = \mathbf{0}</math> has only the trivial solution <math>\mathbf{x} = \mathbf{0}</math>.</li> <li>3. A can be carried to the identity matrix <math>I_n</math> by elementary row operations.</li> <li>4. The system <math>A\mathbf{x} = \mathbf{b}</math> has at least one solution <math>\mathbf{x}</math> for every choice of column <math>\mathbf{b}</math>.</li> <li>5. There exists an <math>n \times n</math> matrix C such that <math>AC = I_n</math>.</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                       | It needs to be square, no empty vars, and the corresponding matrix that reduces it to I is the inverse. NOTE an identity matrix is essentially in RREF without the constants, it represents 1 for each x in $x_1, \dots, x_n$ .                                                                                          |

## Elementary Matrices

| Textbook Ref                                  | Theorem/Definition statement                                                                                                                                                                                     | Notes                                                            |
|-----------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------|
| <b>Definition 2.12</b><br>Elementary Matrices | An $n \times n$ matrix E is called an elementary matrix if it can be obtained from the identity matrix $I_n$ by a single elementary row operation (called the operation corresponding to E). We say that E is of | Elementary row operations in matrix form, convenient application |

|                                                                |                                                                                                                                                                                                                                                      |                                                                                                                                                                                |
|----------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                | type I, II, or III if the operation is of that type: <ul style="list-style-type: none"> <li>I. Interchanging two rows</li> <li>II. Multiplying one row by a non-zero number</li> <li>III. Adding a multiple of one row to a different row</li> </ul> |                                                                                                                                                                                |
| <b>Theorem 2.5.4</b><br>Uniqueness of reduced row-echelon form | If a matrix A is carried to reduced row-echelon matrices R and S by row operations, then $R = S$ .                                                                                                                                                   | If you apply row operations to a matrix and get two different reduced forms, those forms must be equal, no matter how you simplify the matrix, it will return the same result. |

## Linear Transformations

| Textbook Ref                              | Theorem/Definition statement                                                                                                                                                                                                                                                                                                                                                                                                           | Notes                                                                                                                                                                                                                            |
|-------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Definition 2.13</b>                    | <p>A transformation <math>T: \mathbb{R}^n \rightarrow \mathbb{R}^m</math> is called a linear transformation if it satisfies the following two conditions for all vectors <math>\mathbf{x}</math> and <math>\mathbf{y}</math> in <math>\mathbb{R}^n</math> and all scalars <math>a</math>:</p> <p>T1 <math>T(\mathbf{x} + \mathbf{y}) = T(\mathbf{x}) + T(\mathbf{y})</math></p> <p>T2 <math>T(a\mathbf{x}) = aT(\mathbf{x})</math></p> | Bare in mind the behaviour of linear transformations is essentially matrix multiplication, I'll lean more into this idea of a $T()$ function call passing matrix as a param, this describes how the arithmetic can be reordered. |
| <b>Theorem 2.6.1</b><br>Linearity Theorem | <p>If <math>T: \mathbb{R}^n \rightarrow \mathbb{R}^m</math> is a linear transformation, then for each <math>k = 1, 2, \dots</math></p> $T(a_1\mathbf{x}_1 + a_2\mathbf{x}_2 + \dots + a_k\mathbf{x}_k) = a_1T(\mathbf{x}_1) + a_2T(\mathbf{x}_2) + \dots + a_kT(\mathbf{x}_k)$ <p>for all scalars <math>a_i</math> and all vectors <math>\mathbf{x}_i</math> in <math>\mathbb{R}^n</math>.</p>                                         | <p>The effects of multiplying with linearity, scalar multiplication can be applied after a transformation and resolve the same.</p> <p>NOTE continue to actively recognise individual vectors which make up the matrix</p>       |
| <b>Theorem 2.6.4</b><br>Rotations         | <p>The rotation <math>R: \mathbb{R}^2 \rightarrow \mathbb{R}^2</math> is the linear transformation with matrix</p> $\begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$                                                                                                                                                                                                                             | Think about rotation tools in graphic design software, special matrices like this make vector graphics easily editable                                                                                                           |
| <b>Theorem 2.6.5</b><br>Reflections       | <p>Let <math>Q_m</math> denote a reflection transformation with respect to the line <math>y = mx</math>. Then <math>Q_m: \mathbb{R}^2 \rightarrow \mathbb{R}^2</math> is a linear transformation with matrix</p> $\frac{1}{1+m^2} \begin{bmatrix} 1-m^2 & 2m \\ 2m & m^2-1 \end{bmatrix}$                                                                                                                                              | Another graphic design concept, this allows us to easily create mirrored duplicates along an axis/plane                                                                                                                          |
| <b>Theorem 2.6.6</b><br>Projection        | <p>Let <math>P_m</math> be projection on the line <math>y = mx</math>. Then <math>P_m: \mathbb{R}^2 \rightarrow \mathbb{R}^2</math> is a linear transformation with matrix</p>                                                                                                                                                                                                                                                         | This lies at the core of pre-ray-traced graphics and screen-space rendering, determining how 3D objects should appear on a 2D screen from different                                                                              |

|  |                                                                  |        |
|--|------------------------------------------------------------------|--------|
|  | $\frac{1}{1+m^2} \begin{bmatrix} 1 & m \\ m & m^2 \end{bmatrix}$ | angles |
|--|------------------------------------------------------------------|--------|

## Module 3

Module 3 builds on matrix multiplication and RREF reduction to introduce a sequential method for inverting a matrix. This module unpacks how to transform a system linearly through elementary operations, and how elementary matrix products can be used to express reduction/transformation sequences. These are demonstrated with respect to **inverse theorem** and **linear theorem** for maintaining system consistency, clarifying whether the effects can be undone.

### 3.1 Matrix Inverses

Some matrices can be inverted, causing them to induce the inverse-effect of the original matrix transformation. Not all matrices have inverses. If a matrix transformation induces a change, then per **Definition 2.11**, the inverse of that matrix undoes that change. All matrix inverses are unique according to **Theorem 2.4.1**. **Theorem 2.4.5** outlines the 5 conditions of  $n \times n$  matrices according to **inverse theorem**. As a baseline, condition 1 infers that all  $n \times n$  matrices are invertible. Conditions 4 - 5 express further characteristics of invertible matrices, which the **matrix inversion algorithm** operates under. **Inverse theorem** is the conditional framework for how the algorithm works.

#### Summarise the matrix inversion algorithm in words, referring to an example in your evidence or self-assessment

Beginning with the augmented  $[A \mid I]$  matrix, row operations are applied such that the left side is reduced to  $I$ , its corresponding identity. The right side will naturally be brought to  $A^{-1}$ . If the LHS cannot be reduced to  $I$ , the matrix has no inverse, and the algorithm fails. This works within the conditions outlined by **inverse theorem** and characteristics of **inverse properties** for sequential, consistent inversion. The following example demonstrates this with a combination of scalar-mult and scalar-row-add operations. The LHS reduces to RREF as the RHS is carried to  $A^{-1}$ , both with full rank to satisfy **Theorem 2.4.5** (4), so the algorithm succeeds.

**Example 3.1**

3) b)  $2x - 3y = 0$   
 $x - 4y = 1$

$$AX = B \Rightarrow \begin{bmatrix} 2 & -3 & | & 0 & 1 \\ 1 & -4 & | & 0 & 1 \end{bmatrix} \xrightarrow{-2R_2} \begin{bmatrix} 2 & -3 & | & 0 & 1 \\ 0 & 1 & | & -2 & -1 \end{bmatrix} \xrightarrow{+2R_2} \begin{bmatrix} 2 & -1 & | & -4 & -1 \\ 0 & 1 & | & -2 & -1 \end{bmatrix} \xrightarrow{+R_1} \begin{bmatrix} 2 & 0 & | & -6 & -2 \\ 0 & 1 & | & -2 & -1 \end{bmatrix} \xrightarrow{\div 2} \begin{bmatrix} 1 & 0 & | & -3 & -1 \\ 0 & 1 & | & -2 & -1 \end{bmatrix}$$

$$\therefore A^{-1} = \begin{bmatrix} -3 & -1 \\ -2 & -1 \end{bmatrix}$$

$$X = A^{-1}B = \begin{bmatrix} -3 & -1 \\ -2 & -1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} -1 \\ -1 \end{bmatrix}$$

*Retrieve the inverse through row ops*  
*Multiply by the constant vector*

d)  $x + 4y + 2z = 1$   
 $2x + 3y + 3z = -1$   
 $4x + y + 4z = 0$

$$AX = B \Rightarrow \begin{bmatrix} 1 & 4 & 2 & | & 1 & 0 & 0 \\ 2 & 3 & 3 & | & -1 & 0 & 0 \\ 4 & 1 & 4 & | & 0 & 1 & 0 \end{bmatrix} \xrightarrow{-2R_1} \begin{bmatrix} 1 & 4 & 2 & | & 1 & 0 & 0 \\ 0 & -5 & -1 & | & -3 & 0 & 0 \\ 4 & 1 & 4 & | & 0 & 1 & 0 \end{bmatrix} \xrightarrow{-4R_1} \begin{bmatrix} 1 & 4 & 2 & | & 1 & 0 & 0 \\ 0 & -5 & -1 & | & -3 & 0 & 0 \\ 0 & -15 & -4 & | & -4 & 1 & 0 \end{bmatrix} \xrightarrow{+4R_2} \begin{bmatrix} 1 & 4 & 2 & | & 1 & 0 & 0 \\ 0 & -5 & -1 & | & -3 & 0 & 0 \\ 0 & -15 & -4 & | & -4 & 1 & 0 \end{bmatrix} \xrightarrow{+3R_2} \begin{bmatrix} 1 & 4 & 2 & | & 1 & 0 & 0 \\ 0 & -5 & -1 & | & -3 & 0 & 0 \\ 0 & 0 & -1 & | & 5 & 1 & 0 \end{bmatrix} \xrightarrow{\div -1} \begin{bmatrix} 1 & 4 & 2 & | & 1 & 0 & 0 \\ 0 & -5 & -1 & | & -3 & 0 & 0 \\ 0 & 0 & 1 & | & -5 & -1 & 0 \end{bmatrix} \xrightarrow{+R_2} \begin{bmatrix} 1 & 4 & 2 & | & 1 & 0 & 0 \\ 0 & -5 & 0 & | & -8 & -1 & 0 \\ 0 & 0 & 1 & | & -5 & -1 & 0 \end{bmatrix} \xrightarrow{\div -5} \begin{bmatrix} 1 & 4 & 2 & | & 1 & 0 & 0 \\ 0 & 1 & 0 & | & 1.6 & 0.2 & 0 \\ 0 & 0 & 1 & | & -5 & -1 & 0 \end{bmatrix} \xrightarrow{-4R_2} \begin{bmatrix} 1 & 0 & 2 & | & -5.2 & -0.8 & 0 \\ 0 & 1 & 0 & | & 1.6 & 0.2 & 0 \\ 0 & 0 & 1 & | & -5 & -1 & 0 \end{bmatrix} \xrightarrow{-2R_3} \begin{bmatrix} 1 & 0 & 0 & | & 4.8 & 1.2 & 0 \\ 0 & 1 & 0 & | & 1.6 & 0.2 & 0 \\ 0 & 0 & 1 & | & -5 & -1 & 0 \end{bmatrix} \xrightarrow{\div 10} \begin{bmatrix} 1 & 0 & 0 & | & 0.48 & 0.12 & 0 \\ 0 & 1 & 0 & | & 0.16 & 0.02 & 0 \\ 0 & 0 & 1 & | & -0.5 & -0.1 & 0 \end{bmatrix}$$

$$X = A^{-1}B = \begin{bmatrix} 0.48 & 0.12 & 0 \\ 0.16 & 0.02 & 0 \\ -0.5 & -0.1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0.32 \\ 0.02 \\ -0.5 \end{bmatrix}$$

**If the rank of an  $n \times n$  matrix is less than  $n$ , what does this tell us about the inverse matrix?**

**Theorem 2.4.5** clarifies that a matrix with rank less than  $n$  variables cannot be inverted, since the system  $Ax = b$  must have at least one  $x$  for each result in column  $b$ . The **inversion algorithm** will fail in this case. The inverse does not exist for  $n \times n$  matrices that don't hold full rank.

**Provide reasoning as to whether or not a non-square matrix could have an inverse.**

Non-square matrices cannot be inverted. **Definition 2.11** requires  $A$  to be square for  $AA^{-1} = I$  to be consistent. Multiplying a non-square matrix by another cannot yield an  $I$  of compatible dimensions.  $A$  doesn't perfectly reduce to  $I$ , so information gets lost and it breaks system consistency.

## 3.2 Elementary Matrices

**Elementary matrices** represent single elementary row operations applied to the corresponding identity matrix. They're used by the **inverse algorithm** during reduction, and can be composed to represent the series of operations applied to a transformed matrix.

**Theorem 2.5.4** reinforces that any sequence of row operations leads to a unique RREF form, but all are equal, consistent with **linear transformations** inducing linear scaling. Elementary matrices are always invertible, since each operation can be reversed, tying directly to the behaviour outlined in **Theorem 2.4.4** (4), demonstrating inversion over an elementary composition.

## 3.3 Linear Transformations

**Definition 2.13** shows how linear transformations serve as mapping between vector spaces, preserving addition and multiplication linearity, consistent with **linearity theorem**. Each linear transformation can be represented by a matrix, with unique matrices inducing specific effects like **rotations**, **reflections**, and **projections**. In a similar way that the **inversion algorithm** induces a linear reduction on  $A$  to find  $A^{-1}$ , **transformations** induce linear scaling on  $A$  to find the corresponding output. Inversion reverses the order of elementary linear changes induced, so now there's both a way to induce transformation, and a way to reverse/undo it.

**Rotations** and **projections** are both consistent with **inverse properties** and **inverse theorem**. Since rotations and reflections preserve the matrix rank, they're invertible. Projections directly reduce the matrix dimensions, and by **Theorem 2.4.5**, cannot be inverted, due to rank-deficiency with respect to  $n$ .

**What are some strategies for identifying the transformation applied in N3.4.2?**

The type of transformation can be verified by checking whether the  $e_{11}$  and  $e_{22}$  values are equal, indicating a **rotation**. If the  $e_{12}$  and  $e_{21}$  elements are equal, it's likely a **reflection**. This is demonstrated by the following example, which draws on **Theorem 2.6.4** and **Theorem 2.6.5** to clarify that the results are mechanically consistent with their definition structures. The presence of square roots subtly infer angle-calculations, clarified with trig ratios.

113.4.2

a)  $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ ,  $T = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}$  Rotation, if the other two were identical, reflection

returns Q4 for negative vals

$\sin^{-1}\left(\frac{-1}{\sqrt{2}}\right) = -\frac{\pi}{4}$

b) Find a matrix (2x2) that:

- Applies reflection through  $y = -2x$
- Applies a rotation of  $-\frac{\pi}{2}$

$\cos\left(\frac{-\pi}{2}\right) = 0$

$\sin\left(\frac{-\pi}{2}\right) = -1$

$\begin{bmatrix} 0 & -(-1) \\ -1 & 0 \end{bmatrix}$  Rot matrix

$\frac{1}{\sqrt{5}} \begin{pmatrix} -3 & -4 \\ 4 & 3 \end{pmatrix}$  Refl matrix

$R = \frac{1}{1-m^2} \begin{pmatrix} 1-m^2 & 2m \\ 2m & -(1-m^2) \end{pmatrix}$  Reflection

$\begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \begin{bmatrix} \frac{3}{5} & \frac{4}{5} \\ \frac{4}{5} & \frac{3}{5} \end{bmatrix} = \begin{bmatrix} \frac{4}{5} & \frac{3}{5} \\ \frac{3}{5} & \frac{4}{5} \end{bmatrix}$

This draws on Theorem 2.6.4, 2.6.5 to determine the type of linear transformation

Rotation

Reflection

Module 3 expresses transformations and inversion as two opposing, tightly-coupled mechanisms. Inverse matrices reverse the order of operations induced by an elementary composition, and by extension, the effects induced by the overall transformation. This scopes row ops within elementary matrices, and their sequence within composition. If a linear system is subject to linear transformation from another matrix, it's only possible to retrieve the original system if the matrix can be inverted, reversing the elementary operations the transformation is composed of. If a transformation is considered, the existence of a way to reverse it needs to be clarified, an extremely important condition for consistent linear systems.

## Reflecting on the content:

This module was a good point for tying together Modules 1, 2, and composition/transformations. I had a few problems getting my calculator configured for working in fractions for the trigonometry aspects, but the most important concepts were very intuitive to work with. I had to work with some rephrasing to really get my head around the composition annotations, but I feel more confident now that I've got 3 clear stages of progression. The hierarchy established in this module will help me scope the upcoming modules with better context now that these core building blocks are established. Overall, a very important module for using all module 1 and 2 concepts in context.

## References

### Summary and Notes

[1] D. Margalit and J. Rabinoff, "Matrix Transformations," Interactive Linear Algebra, LibreTexts, [Online]. Available: [https://math.libretexts.org/Bookshelves/Linear\\_Algebra/Interactive\\_Linear\\_Algebra\\_\(Margalit\\_and\\_Rabinoff\)/03%3A\\_Linear\\_Transformations\\_and\\_Matrix\\_Algebra/3.01%3A\\_Matrix\\_Transformations](https://math.libretexts.org/Bookshelves/Linear_Algebra/Interactive_Linear_Algebra_(Margalit_and_Rabinoff)/03%3A_Linear_Transformations_and_Matrix_Algebra/3.01%3A_Matrix_Transformations). [Accessed: Jul. 30, 2025].

[2] D. Margalit and J. Rabinoff, "Matrix Inverses," Interactive Linear Algebra, LibreTexts, [Online]. Available: [https://math.libretexts.org/Bookshelves/Linear\\_Algebra/Interactive\\_Linear\\_Algebra\\_\(Margalit\\_and\\_Rabinoff\)/03%3A\\_Linear\\_Transformations\\_and\\_Matrix\\_Algebra/3.05%3A\\_Matrix\\_Inverses](https://math.libretexts.org/Bookshelves/Linear_Algebra/Interactive_Linear_Algebra_(Margalit_and_Rabinoff)/03%3A_Linear_Transformations_and_Matrix_Algebra/3.05%3A_Matrix_Inverses). [Accessed: Jul. 30, 2025].

[3] A. Doerr and K. Levasseur, "Matrix Inversion," Applied Discrete Structures, LibreTexts, [Online]. Available: [https://math.libretexts.org/Bookshelves/Combinatorics\\_and\\_Discrete\\_Mathematics/Applied\\_Discrete\\_Structures\\_\(Doerr\\_and\\_Levasseur\)/12%3A\\_More\\_Matrix\\_Algebra/12.02%3A\\_Matrix\\_Inversion](https://math.libretexts.org/Bookshelves/Combinatorics_and_Discrete_Mathematics/Applied_Discrete_Structures_(Doerr_and_Levasseur)/12%3A_More_Matrix_Algebra/12.02%3A_Matrix_Inversion). [Accessed: Jul. 30, 2025].

### Calculations/Verification

[4] Apple Inc., "Calculator Infinity," App Store. [Online]. Available: <https://apps.apple.com/us/app/calculator/id458535809>.